

Entropy

Throughout, $\log = \log_2$, the unit is the **bit**, and $0 \log 0 = 0$. One problem carries the first half: sending instructions to a robot when every bit costs money. One picture carries the second half: the two-dice table of the Conditional Expectation lecture.

1. **Codes** Prefix-free codes are binary trees; a codeword of length ℓ consumes $2^{-\ell}$ of the tree (Kraft).
2. **Information** $-\log p$ The ideal codeword length: the number of halvings from 1 down to p .
3. **Entropy** H Expected information; the area under stacked bars.
4. **Source coding theorem** $H \leq \text{best average code length} < H + 1$.
5. **Cross-entropy, relative entropy** The cost of coding for the wrong distribution, and the waste. Gibbs' inequality.
6. **Why the loss is a logarithm** Honesty forces $-\log q$; cross-entropy loss; prediction is compression.
7. **Wordle** A guess is a partition; its entropy is its expected information.
8. **Joint, conditional, mutual** Row entropies of the two-dice table; chain rule; $I(X; Y)$.
9. **Models of text** Markov chains, entropy rate, English at about 1 bit per letter, perplexity, ciphers.

Motivation: A Robot on a Distant Moon

A rover receives a stream of instructions, each one of up, down, left, right, independent of the others, with probabilities

$$\mathbb{P}(\text{up}) = \frac{1}{2}, \quad \mathbb{P}(\text{down}) = \frac{1}{4}, \quad \mathbb{P}(\text{left}) = \frac{1}{8}, \quad \mathbb{P}(\text{right}) = \frac{1}{8}.$$

Every bit sent costs money.

Head-on: a fixed-length code

up $\mapsto$ 00, down $\mapsto$ 01, left $\mapsto$ 10, right $\mapsto$ 11: cost 2 bits per instruction, whatever the probabilities.

Questions we cannot yet answer

1. Can a cleverer encoding average fewer than 2 bits per instruction?
2. If so, how low can the average go? Is there a floor, and what is it?
3. The rover's mission changes and right becomes the common instruction, but the code installed on board stays. What does that cost?

Pre-lecture viewing: "Reinventing Entropy" (3Blue1Brown, 32 min),

<https://www.youtube.com/watch?v=l6DKRf-fAAM>

The Setting: A Source of Symbols

Fix a probability space $(\Omega, \mathcal{F}, \mathbb{P})$; events are the elements of $\mathcal{F}$.

- S is a finite set of **symbols**, $|S| = n$. Label them $1, \dots, n$ if you like, so that $X: \Omega \rightarrow S$ is a random variable in the sense of the Refresher: $\{X = x\} \in \mathcal{F}$ for every $x \in S$.
- Shorthand, used throughout: $\{X \in A\} := \{\omega \in \Omega : X(\omega) \in A\}$ and $\mathbb{P}(X \in A) := \mathbb{P}(\{X \in A\})$.
- The **pmf** of X is $p_X: S \rightarrow [0, 1]$, $p_X(x) = \mathbb{P}(X = x)$; its **support** is $S_X = \{x \in S : p_X(x) > 0\}$.

Definition (source)

A **source** is a sequence of random variables $X_1, X_2, \dots: \Omega \rightarrow S$, independent and all with pmf p_X . Independence, written $X_i \perp\!\!\!\perp X_j$, means $\mathbb{P}(X_i \in A, X_j \in B) = \mathbb{P}(X_i \in A)\mathbb{P}(X_j \in B)$ for all $A, B \subseteq S$; for the whole sequence,

$$\mathbb{P}(X_1 = x_1, \dots, X_m = x_m) = p_X(x_1) \cdots p_X(x_m) \quad \text{for every } m \text{ and all } x_1, \dots, x_m \in S.$$

A source is a die with n faces, rolled again and again; we are charged for reporting the outcomes.

- The rover: $S = \{\text{up}, \text{down}, \text{left}, \text{right}\}$, $p_X = (\frac{1}{2}, \frac{1}{4}, \frac{1}{8}, \frac{1}{8})$, $S_X = S$, and X_i is the i -th instruction.

Binary Code

Let $\{0, 1\}^+$ be the set of non-empty finite strings of bits: 0, 1, 00, 01, ...

Definition

A **binary code** for S is an injective function $c: S \rightarrow \{0, 1\}^+$; $c(x)$ is the **codeword** of x . Its **length function** is $\ell_c: S \rightarrow \{1, 2, 3, \dots\}$, $\ell_c(x)$ = the number of bits in $c(x)$, and its **average length** is

$$L(c) = \mathbb{E}[\ell_c(X)] = \sum_{x \in S} p_X(x) \ell_c(x) \quad (\text{bits per symbol}).$$

The code is **prefix-free** if for $x \neq x'$ the string $c(x)$ is not an initial segment of $c(x')$.

Each symbol gets a bit string; the average length is what one symbol costs on average; prefix-free means the receiver knows where every codeword ends without a separator.

The rover, coded

c : up $\mapsto$ 0, down $\mapsto$ 10, left $\mapsto$ 110, right $\mapsto$ 111. Prefix-free: no codeword begins another.

$$L(c) = \frac{1}{2} \cdot 1 + \frac{1}{4} \cdot 2 + \frac{1}{8} \cdot 3 + \frac{1}{8} \cdot 3 = 1.75 \text{ bits per instruction, against 2.}$$

The stream 0 110 100 111 is decoded left to right with no ambiguity: up, left, down, up, right.

Prefix-Free Codes Are Binary Trees

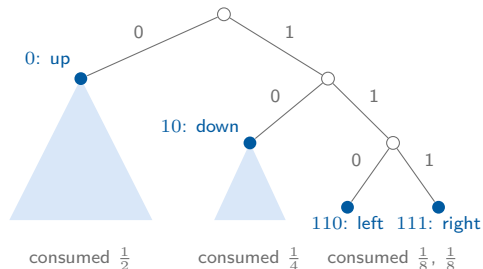
Read a codeword as a path from the root: 0 = left, 1 = right. Then $c(x)$ is a prefix of $c(x')$ exactly when the node of $c(x)$ lies on the path to $c(x')$; so a code is prefix-free exactly when **every codeword is a leaf**.

Choosing 0 as a codeword consumes the whole left subtree: half of all strings, of every length, can no longer be codewords. Choosing 10 consumes a quarter; 110 and 111 an eighth each:

$$\frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \frac{1}{8} = 1.$$

A codeword of length ℓ uses up the fraction $2^{-\ell}$ of the space of codewords. In the rover's code the fractions used up are exactly the probabilities.

Not a code: up $\mapsto$ 0, down $\mapsto$ 1, left $\mapsto$ 01, right $\mapsto$ 10. Injective, 1.5 bits on average, useless: 01 reads as up down or as left. Here 0 and 1 are not leaves.



0: $\frac{1}{2}$	10: $\frac{1}{4}$	110	111
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the space of codewords, total 1

Kraft's Inequality

Theorem (Kraft, 1949)

If a prefix-free code for $S = \{x_1, \dots, x_n\}$ has lengths $\ell_1, \dots, \ell_n$, then

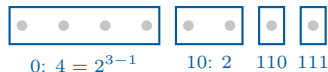
$$\sum_{i=1}^n 2^{-\ell_i} \leq 1.$$

Conversely, for any positive integers $\ell_1, \dots, \ell_n$ with $\sum_i 2^{-\ell_i} \leq 1$ there is a prefix-free code with exactly these lengths.

The fractions of codeword space consumed by the codewords must fit inside 1; and any lengths that fit can be realised.

- Rover: $2^{-1} + 2^{-2} + 2^{-3} + 2^{-3} = 1$. The tree is full.
- Lengths (1, 2, 2, 3) are impossible for any four symbols: $\frac{1}{2} + \frac{1}{4} + \frac{1}{4} + \frac{1}{8} = \frac{9}{8} > 1$.
- Lengths (2, 2, 2, 2): $4 \cdot \frac{1}{4} = 1$, the fixed-length code. Lengths (1, 2, 3, 4): $\frac{15}{16} < 1$, realisable.

the $2^L = 8$ strings of length $L = 3$



Beneath a codeword of length ℓ lie $2^{L-\ell}$ strings of length L ; prefix-free makes these sets disjoint.

Kraft's Inequality: Proof

Proof of the inequality

Let $L = \max_i \ell_i$ and grow the tree to depth L : it has 2^L leaves. Beneath the node of a codeword of length ℓ_i lie $2^{L-\ell_i}$ of them. Prefix-free means no codeword lies beneath another, so these sets of leaves are disjoint, and $\sum_i 2^{L-\ell_i} \leq 2^L$. Divide by 2^L . $\square$

Proof of the converse

Order $\ell_1 \leq \dots \leq \ell_n$ and choose codewords in turn: for x_i take the leftmost depth- ℓ_i node not beneath a chosen codeword. The codewords chosen so far cover the leftmost $\sum_{j < i} 2^{-\ell_j} < 1$ of codeword space, in blocks whose sizes $2^{-\ell_j}$ are multiples of $2^{-\ell_i}$ (as $\ell_j \leq \ell_i$); so a free depth- ℓ_i node exists immediately to their right. Repeat for $i = 1, \dots, n$. $\square$

Each codeword owns a disjoint block of the 2^L strings of the longest length; there are only 2^L to own.

- The converse at work, lengths (1, 2, 3, 4): take 0; then 00, 01 lie beneath 0, so take 10; then 110; then 1110. Consumed $\frac{15}{16}$; the node 1111 stays free.

Exercise

1. Decode 1100111010 with the rover's code. Then encode right, up, up, down and count the bits.
2. Which of the length profiles $(1, 2, 2, 3)$, $(1, 2, 3, 4)$, $(2, 2, 2, 2)$, $(1, 3, 3, 3)$ can be the lengths of a prefix-free code for four symbols? For each admissible one, write down such a code and its average length under $(\frac{1}{2}, \frac{1}{4}, \frac{1}{8}, \frac{1}{8})$.
3. $p_X = (\frac{1}{2}, \frac{1}{4}, \frac{1}{4})$. Design a prefix-free code with average length 1.5 and draw its tree.
4. Is $\{0, 01, 011, 0111\}$ prefix-free? Is every stream built from it nevertheless decodable? Compute $\sum 2^{-\ell}$ and give a prefix-free code with the same lengths.
5. A fair die. Show that no prefix-free code has all six lengths equal to 2; find one with lengths $(2, 2, 3, 3, 3, 3)$ and its average length. Can a prefix-free code average 2.5 bits per roll? (Return to this after Section 4.)

Kraft's inequality turns "is there a prefix-free code with these lengths?" into an arithmetic check.

Information

In the rover's code, up costs 1 bit, down 2, left and right 3; and $\frac{1}{2} = 2^{-1}$, $\frac{1}{4} = 2^{-2}$, $\frac{1}{8} = 2^{-3}$. The codeword length is the number of times 1 must be halved to reach the probability.

Definition

The **information** (surprisal) of an event is the function $I: \{A \in \mathcal{F} : \mathbb{P}(A) > 0\} \rightarrow [0, \infty)$,

$$I(A) = \log \frac{1}{\mathbb{P}(A)} = -\log \mathbb{P}(A) \quad \text{bits.}$$

For a random variable X with pmf p_X , the **information function** is $I_X: S_X \rightarrow [0, \infty)$, $I_X(x) = -\log p_X(x)$; composing it with X gives the random variable $I_X(X): \Omega \rightarrow [0, \infty)$, $\omega \mapsto -\log p_X(X(\omega))$.

$I(A)$ is the number of halvings from certainty down to $\mathbb{P}(A)$: the number of fair coin tosses that would be exactly as surprising as A .

Examples

Rover: $I_X(\text{up}) = 1$, $I_X(\text{down}) = 2$, $I_X(\text{left}) = I_X(\text{right}) = 3$ bits, the codeword lengths. A six: $\log 6 = 2.585$ bits. Double six: $\log 36 = 5.170 = 2.585 + 2.585$ bits. A specific sequence of 20 coin tosses: 20 bits, whatever it is.

Why a Logarithm: Additivity Under Independence

Wanted: $f: (0, 1] \rightarrow [0, \infty)$, “the information of an event of probability p ”, such that (i) f is non-increasing: a rarer event is at least as informative; (ii) $f(pq) = f(p) + f(q)$: for $A \perp\!\!\!\perp B$ learning both is learning each in turn, so $I(A \cap B) = I(A) + I(B)$ while $\mathbb{P}(A \cap B) = \mathbb{P}(A)\mathbb{P}(B)$; (iii) $f(\frac{1}{2}) = 1$: a fair coin toss is one bit.

Theorem

The only such function is $f(p) = -\log p$.

Proof

From (ii), $f(1) = f(1 \cdot 1) = 2f(1)$, so $f(1) = 0$; and $f(2^{-k}) = k f(\frac{1}{2}) = k$ for $k = 0, 1, 2, \dots$. For rational $r = m/n$, applying (ii) n times gives $n f(2^{-m/n}) = f(2^{-m}) = m$, so $f(2^{-r}) = r$: $f(p) = -\log p$ whenever $-\log p$ is rational. For any other p , take rationals $r < -\log p < r'$ as close as we like; then $2^{-r'} < p < 2^{-r}$ and (i) gives $r \leq f(p) \leq r'$. $\square$

Independence multiplies probabilities; information must add; only a logarithm turns products into sums, and calling a coin toss one bit fixes the base at 2.

- With $\ln$ instead of $\log$ the unit is the *nat*; $1 \text{ nat} = 1/\ln 2 = 1.443$ bits. Nothing else changes.

What Information Is Not

not. . .	the fact	one number
. . . about the pattern	$\mathbb{P}(A) = \mathbb{P}(B) \Rightarrow I(A) = I(B)$	HHHHHHHHHH and HTHHTTHTHT: 10 bits each.
. . . the length you happened to use	$I_X(x)$ is the <i>ideal</i> length; $\ell_c(x)$ is a choice	fixed-length code: $\ell_c(\text{up}) = 2 \neq I_X(\text{up}) = 1$; rover's code: $\ell_c = I_X$, and it is optimal (Section 4).
. . . one object	$I(A) \in [0, \infty)$; $I_X: S_X \rightarrow [0, \infty)$; $I_X(X): \Omega \rightarrow [0, \infty)$	a number for an event; a function of the value; a random variable, with mean $H(X)$ (Section 3).
. . . additive without independence	$I(A \cap B) = I(A) + I(B)$ iff $\mathbb{P}(A \cap B) = \mathbb{P}(A)\mathbb{P}(B)$	two dice, Y the first, X the sum: $I(\{Y = 1\} \cap \{X = 2\}) = \log 36 =$ 5.170, but $I(\{Y = 1\}) + I(\{X =$ 2}) = 2.585 + 5.170.
. . . negative, and zero only for the certain	$I(A) \geq 0$, with $I(A) = 0$ iff $\mathbb{P}(A) = 1$	$I(\Omega) = 0$; $I(\text{a six}) = 2.585$.

Information is a property of the probability of what you learned, not of what it looked like or how it was transmitted.

Back to the Rover: The Coded Stream Is Fair Coin Flips

Send instructions with the code 0, 10, 110, 111 and look at the stream of bits, one at a time.

Each bit is a fair coin flip

$$\mathbb{P}(\text{1st bit} = 0) = \mathbb{P}(\text{up}) = \frac{1}{2},$$

$$\mathbb{P}(\text{2nd bit} = 0 \mid \text{1st bit} = 1) = \frac{\mathbb{P}(\text{down})}{\mathbb{P}(\text{down}) + \mathbb{P}(\text{left}) + \mathbb{P}(\text{right})} = \frac{1/4}{1/2} = \frac{1}{2},$$

$$\mathbb{P}(\text{3rd bit} = 0 \mid \text{first two} = 11) = \frac{\mathbb{P}(\text{left})}{\mathbb{P}(\text{left}) + \mathbb{P}(\text{right})} = \frac{1/8}{1/4} = \frac{1}{2}.$$

Every internal node of the tree splits the probability beneath it in half, and after a codeword ends the next instruction is independent of everything before; so every bit of the stream is 0 with probability $\frac{1}{2}$, independently of all earlier bits.

A perfectly compressed stream is indistinguishable from noise; and noise cannot be compressed, since a fair coin flip costs one bit and there is nothing left to predict.

- Compressing the rover's stream a second time gains nothing. Compression removes predictability; what remains is exactly unpredictable.

Exercise

1. How many bits of information are in HHHH? In a royal flush, $\mathbb{P} = 4/2\,598\,960$? An event has 10 bits of information: what is its probability?
2. A fair die is rolled. How much information do you receive on being told that it is even? That it is at least 5? That it is a 6?
3. Two dice, Y the first, X the sum. Compute $I(\{Y = 1\})$, $I(\{X = 7\})$, $I(\{Y = 1\} \cap \{X = 7\})$ and explain the relation between the three. Repeat with $\{X = 2\}$ in place of $\{X = 7\}$ and explain the difference.
4. Using only $f(pq) = f(p) + f(q)$ and $f(\frac{1}{2}) = 1$, show that $f(1) = 0$, $f(\frac{1}{8}) = 3$ and $f(1/\sqrt{2}) = \frac{1}{2}$.
5. For which pmfs does some prefix-free code produce a stream in which every bit is a fair coin flip independent of the earlier bits? Prove your condition is necessary.

Information converts a probability into a count of fair coin tosses, and under independence the counts add.

Entropy

What does one instruction cost on average at the ideal lengths? Weight each $I_X(x)$ by its probability.

Definition

The **entropy** of X is the expected information of its value,

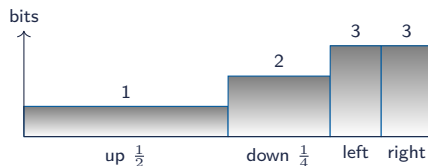
$$H(X) = \mathbb{E}[I_X(X)] = \sum_{x \in S_X} p_X(x) \log \frac{1}{p_X(x)} \quad \text{bits.}$$

It depends on X only through its pmf, so for a pmf p we also write $H(p)$: $H: \Delta_n \rightarrow [0, \log n]$, $\Delta_n = \{p \in [0, 1]^n : \sum_i p_i = 1\}$ (upper bound: Section 5).

The average number of bits per symbol when every symbol gets its ideal codeword length; equivalently, the average surprise on seeing the value of X .

The rover

$H(\frac{1}{2}, \frac{1}{4}, \frac{1}{8}, \frac{1}{8}) = \frac{1}{2} \cdot 1 + \frac{1}{4} \cdot 2 + \frac{1}{8} \cdot 3 + \frac{1}{8} \cdot 3 = 1.75$ bits:
exactly the average length of the code 0, 10, 110, 111.



Bar i : width p_i , height $\log(1/p_i)$, area $p_i \log(1/p_i)$. Total area = $H(p)$.

What Entropy Is Not

not. . .	the fact	one number
. . . a random variable	$I_X(X): \Omega \rightarrow [0, \infty)$ is the random variable; $H(X) = \mathbb{E}[I_X(X)] \in [0, \infty)$ is its mean	rover: $I_X(X) \in \{1, 2, 3\}$; $H(X) = 1.75$.
. . . about the values	$H(g(X)) = H(X)$ for injective g ; entropy is not variance	fair coin on $\{0, 1\}$ or on $\{0, 1000\}$: $H = 1$ bit both times; variances 0.25 and 250 000.
. . . negative	$p \log \frac{1}{p} \geq 0$, $= 0$ iff $p \in \{0, 1\}$; so $H(X) \geq 0$, $= 0$ iff X is constant	$H(1, 0, 0, 0) = 0$.
. . . more than $\log n$	$H(X) \leq \log n$ for n values, $=$ iff uniform (Section 5)	fair die $\log 6 = 2.585$; fair coin 1.
. . . the number of values	$H(X) < \log n$ unless X is uniform	sum of two dice: 11 values, $\log 11 = 3.459$, but $H = 3.274$ (two slides on).
. . . the cost of one symbol	$I_X(x)$ is that; $H(X)$ is the average over a long stream	rover: left costs 3 bits; the stream costs 1.75 per instruction.

Entropy measures how spread out the probabilities are, not how spread out the values are.

Binary Entropy Function (h)

Definition

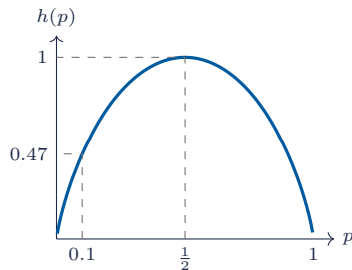
The **binary entropy function** is

$$h: [0, 1] \rightarrow [0, 1], \quad h(p) = p \log \frac{1}{p} + (1-p) \log \frac{1}{1-p},$$

the entropy of a coin with $\mathbb{P}(H) = p$, i.e. of the pmf $(p, 1-p)$.

One curve covers every two-outcome random variable, since its entropy depends only on the probability of one outcome.

- $h(0) = h(1) = 0$; maximum $h(\frac{1}{2}) = 1$; symmetric, $h(p) = h(1-p)$; concave, with infinite slope at 0 and 1.
- A coin showing heads 99% of the time still has $h(0.01) = 0.081$ bits: the rare tail costs $\log 100 = 6.64$ bits when it comes.



Values that recur below: $h(0.1) = 0.469$,
 $h(0.2) = 0.722$, $h(\frac{1}{4}) = 0.811$,
 $h(0.3) = 0.881$.

Example: The Sum of Two Dice Costs 3.27 Bits per Roll

Two fair dice, Y the first, Z the second, $X = Y + Z$: the two-dice table. What does it cost per roll to report the sum?

- Head-on: 11 values, so 4 bits with a fixed-length code; $\log 11 = 3.459$ if fractional lengths were allowed. Both ignore that 7 is six times as likely as 2.
- The pmf: $p_X(x) = n_x/36$, where n_x is the number of cells on the anti-diagonal $\{X = x\}$:
 $n_x = 1, 2, 3, 4, 5, 6, 5, 4, 3, 2, 1$ for $x = 2, \dots, 12$.
- For comparison: one die, $H(Y) = \log 6 = 2.585$; the pair (Y, Z) , uniform on 36 cells, $\log 36 = 5.170$.

$y \setminus z$	1	2	3	4	5	6
1	2	3	4	5	6	7
2	3	4	5	6	7	8
3	4	5	6	7	8	9
4	5	6	7	8	9	10
5	6	7	8	9	10	11
6	7	8	9	10	11	12

$\{X = 7\}$: six cells. $\{X = 2\}$: one cell.

$$H(X) = \sum_{x=2}^{12} \frac{n_x}{36} \log \frac{36}{n_x} = 2 \left[\frac{1}{36} \log 36 + \frac{2}{36} \log 18 + \frac{3}{36} \log 12 + \frac{4}{36} \log 9 + \frac{5}{36} \log \frac{36}{5} \right] + \frac{6}{36} \log 6 = 3.274.$$

More than one die, less than the pair it is computed from: a function of the outcome can only lose information (Section 8).

Exercise

1. Draw the area picture for the uniform pmf $(\frac{1}{4}, \frac{1}{4}, \frac{1}{4}, \frac{1}{4})$ and read off H . Do the same for $(\frac{1}{2}, \frac{1}{4}, \frac{1}{4})$.
2. X a fair die, $W = \mathbf{1}_{\{X \text{ even}\}}$ and $V = \min(X, 3)$. Compute $H(W)$ and $H(V)$.
3. Which has more entropy, the sum of two dice or the maximum of two dice? Compute both. Which is cheaper to transmit?
4. Prove from the definition that $H(X) \geq 0$, with equality if and only if X is constant.
5. A source emits A with probability 0.9 and B with probability 0.1. Its entropy is $h(0.9) = 0.469$ bits per symbol, yet every codeword has at least one bit, so no code for single symbols averages under 1 bit per symbol. Find a way to get below 1. (Hint: code two symbols at a time.)

Entropy is the average ideal codeword length, so it is the number to compute before choosing what to transmit.

The Source Coding Theorem

Let $L^* = \min\{L(c) : c \text{ a prefix-free code for } S\}$, the best achievable average length.

Theorem (Shannon, 1948)

$$H(X) \leq L^* < H(X) + 1.$$

The left inequality is an equality if and only if every $p_X(x)$ is a power of 2, in which case the optimal code has $\ell(x) = -\log p_X(x) = I_X(x)$ for every $x \in S_X$.

Entropy is the floor on the number of bits per symbol that any prefix-free code can average, and a code within one bit of the floor always exists.

Back to the rover

$H = 1.75$ and the code 0, 10, 110, 111 has $L = 1.75$: it is optimal. No prefix-free code averages fewer than 1.75 bits per instruction. That answers questions 1 and 2 of the opening slide.

- The bound $L^* \geq H$ is what makes entropy *the* measure of information: not a convention, but the price no code can beat.
- The proof of the lower bound needs one inequality, used again for relative entropy in Section 5. It comes first.

Gibbs' Inequality (Lemma)

Lemma (Gibbs' inequality, general form)

Let p be a pmf on S with support $S_p = \{x \in S : p(x) > 0\}$ (for $p = p_X$ this is S_X), and let $a: S \rightarrow [0, \infty)$ satisfy $a > 0$ on S_p and $\sum_{x \in S} a(x) \leq 1$. Then

$$\sum_{x \in S_p} p(x) \log \frac{1}{a(x)} \geq \sum_{x \in S_p} p(x) \log \frac{1}{p(x)}, \text{ with equality if and only if } a(x) = p(x) \text{ for every } x \in S.$$

Weights from p , lengths $\log(1/a)$ from anything that fits under Kraft: the average is smallest with p 's own lengths.

Proof

For $t > 0$, $\ln t \leq t - 1$, with equality only at $t = 1$. Apply it at $t = a(x)/p(x)$ for $x \in S_p$:

$$\sum_{x \in S_p} p(x) \ln \frac{a(x)}{p(x)} \leq \sum_{x \in S_p} p(x) \left(\frac{a(x)}{p(x)} - 1 \right) = \sum_{x \in S_p} a(x) - 1 \leq \sum_{x \in S} a(x) - 1 \leq 0.$$

Divide by $\ln 2$. Equality forces $a = p$ on S_p (first inequality) and $\sum_{S_p} a = 1$ (second and third), so $a = 0 = p$ off S_p . □

Source Coding Theorem: The Lower Bound

Proof that $H(X) \leq L(c)$ for every prefix-free code c

Put $a(x) = 2^{-\ell_c(x)} > 0$ for $x \in S$. By Kraft, $\sum_x a(x) \leq 1$; and $\log \frac{1}{a(x)} = \ell_c(x)$. Gibbs' lemma with $p = p_X$ gives

$$L(c) = \sum_{x \in S_X} p_X(x) \ell_c(x) = \sum_{x \in S_X} p_X(x) \log \frac{1}{a(x)} \geq \sum_{x \in S_X} p_X(x) \log \frac{1}{p_X(x)} = H(X),$$

with equality if and only if $2^{-\ell_c(x)} = p_X(x)$ for all $x \in S$: the probabilities are powers of 2 and the lengths are the informations. $\square$

The lengths of any prefix-free code define “probabilities” $2^{-\ell(x)}$; coding with lengths made for those costs at least as much as coding with lengths made for the truth.

The rover, in numbers

Any prefix-free code for four symbols has $a = (2^{-\ell_1}, \dots, 2^{-\ell_4})$ with $\sum a \leq 1$, so $L(c) = \frac{1}{2}\ell_1 + \frac{1}{4}\ell_2 + \frac{1}{8}\ell_3 + \frac{1}{8}\ell_4 = \sum_x p_X(x) \log \frac{1}{a(x)} \geq \sum_x p_X(x) \log \frac{1}{p_X(x)} = 1.75$. The lengths $(1, 2, 3, 3)$ give $a = p_X$ exactly, hence equality; no other lengths do.

Source Coding Theorem: The Upper Bound

Proof that $L^* < H(X) + 1$ (Shannon's construction)

Take $S_X = S$ (a symbol of probability 0 never occurs and needs no codeword). Round each ideal length up: $\ell(x) = \lceil \log \frac{1}{p_X(x)} \rceil$. Then $2^{-\ell(x)} \leq p_X(x)$, so $\sum_x 2^{-\ell(x)} \leq \sum_x p_X(x) = 1$, and Kraft supplies a prefix-free code c with these lengths. Its average length is

$$L(c) = \sum_x p_X(x) \left\lceil \log \frac{1}{p_X(x)} \right\rceil < \sum_x p_X(x) \left(\log \frac{1}{p_X(x)} + 1 \right) = H(X) + 1. \quad \square$$

Rounding every ideal length up to a whole number wastes less than one bit per symbol.

Example: $p = (0.4, 0.3, 0.2, 0.1)$

Ideal lengths 1.32, 1.74, 2.32, 3.32; rounded up: 2, 2, 3, 4; Kraft sum $\frac{1}{4} + \frac{1}{4} + \frac{1}{8} + \frac{1}{16} = \frac{11}{16}$; code 00, 01, 100, 1010 by the converse of Kraft.

$$L = 0.4 \cdot 2 + 0.3 \cdot 2 + 0.2 \cdot 3 + 0.1 \cdot 4 = 2.4 \text{ bits}, \quad H = 1.846, \quad L < H + 1 = 2.846.$$

Shannon's code is not the best one: lengths (1, 2, 3, 3) also satisfy Kraft and give $L = 1.9$. The theorem only promises a code within one bit of H .

Entropy Adds Under Independence

Theorem

If $X \perp\!\!\!\perp Y$ then the pair (X, Y) , with pmf $p_{X,Y}(x, y) = \mathbb{P}(X = x, Y = y)$, has entropy $H(X, Y) = H(X) + H(Y)$.

Proof

$p_{X,Y}(x, y) = p_X(x)p_Y(y)$, so $\log \frac{1}{p_{X,Y}(x,y)} = \log \frac{1}{p_X(x)} + \log \frac{1}{p_Y(y)}$. Multiply by $p_{X,Y}(x, y)$ and sum over (x, y) . In the first term, $\sum_y p_{X,Y}(x, y) = p_X(x)$, leaving $\sum_x p_X(x) \log \frac{1}{p_X(x)} = H(X)$; likewise the second term gives $H(Y)$. $\square$

Independent symbols carry independent surprises, and expectations add.

Examples

Two dice, Y and Z : $H(Y, Z) = \log 6 + \log 6 = \log 36 = 5.170$, the uniform pmf on the 36 cells. k source symbols: $H(X_1, \dots, X_k) = kH(X)$, by induction.

- The converse fails for the same reason it fails for expectations: Section 8 shows $H(X, Y) \leq H(X) + H(Y)$ always, with equality only under independence.

Closing the Gap: Code Blocks of Symbols

Exercise 5 of Section 3: the A/B source has $h(0.9) = 0.469$ bits per symbol, yet every codeword has at least one bit. Code several symbols at once.

Theorem (entropy is exactly the limit)

Let L_k^* be the best average length of a prefix-free code for blocks $(X_1, \dots, X_k)$ of k source symbols. Then

$$H(X) \leq \frac{L_k^*}{k} < H(X) + \frac{1}{k}.$$

Proof

The block is a symbol of the alphabet S^k with entropy $kH(X)$ (previous slide). Apply the source coding theorem to it: $kH(X) \leq L_k^* < kH(X) + 1$. Divide by k . □

Per symbol, the unavoidable rounding loss shrinks to $1/k$; the floor $H(X)$ never moves.

The A/B source

Pairs have pmf $(0.81, 0.09, 0.09, 0.01)$. The code 0, 10, 110, 111 costs $0.81 + 0.18 + 0.27 + 0.03 = 1.29$ bits per pair, i.e. 0.645 bits per symbol, already below 1; the floor is 0.469.

Exercise

1. A fair die: Shannon's lengths are $\lceil \log 6 \rceil = 3$ for every face, $L = 3$. The code with lengths $(2, 2, 3, 3, 3, 3)$ averages 2.667. Can any prefix-free code average 2.5? Answer exercise 5 of Section 1.
2. Write out, for the rover's four probabilities, the three lines of the lower-bound proof with numbers, and conclude that no prefix-free code beats 1.75 bits per instruction.
3. The theorem says $L^* < h(0.9) + 1 = 1.469$ for the A/B source, and single-symbol codes cannot go below 1. Why is there no contradiction? Compute the Shannon lengths for pairs of symbols and the resulting bits per symbol.
4. For the A/B source, how many symbols per block guarantee a Shannon code under 0.6 bits per symbol? Why does the same argument fail to get under $h(0.9) = 0.469$?
5. If every $p_X(x)$ is a power of 2, show that the Shannon code has $L = H(X)$, and that every bit it emits is a fair coin flip independent of the earlier bits.

The source coding theorem is a floor no code can beat and a ceiling within one bit of it, and coding blocks closes the gap.

Cross-Entropy

The rover's mission changes: right is now the common instruction, $p = (\frac{1}{8}, \frac{1}{8}, \frac{1}{4}, \frac{1}{2})$ for (up, down, left, right). The code 0, 10, 110, 111 stays installed.

$$\text{cost} = \frac{1}{8} \cdot 1 + \frac{1}{8} \cdot 2 + \frac{1}{4} \cdot 3 + \frac{1}{2} \cdot 3 = 2.625 \text{ bits per instruction,}$$

worse than the 2 bits of the fixed-length code, while the new entropy is still $H(p) = 1.75$.

Definition

Let P, Q be pmfs on S , with $S_P = \{x : P(x) > 0\}$. The **cross-entropy** of P with Q is

$$H(P, Q) = \sum_{x \in S_P} P(x) \log \frac{1}{Q(x)} \quad \text{bits, with } \log \frac{1}{0} = \infty,$$

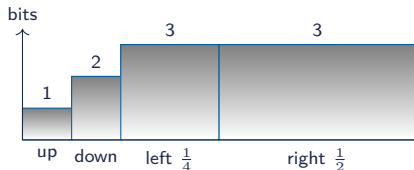
a function $H: \Delta_n \times \Delta_n \rightarrow [0, \infty]$. The first argument P is the *truth*: it supplies the weights. The second, Q , is the *code* or *model*: it supplies the lengths $\log(1/Q(x))$. Notation: H of two *pmfs* is cross-entropy; $H(X, Y)$ of two *random variables* is the entropy of the pair (Section 4, Section 8).

The average bits per symbol when the code was built for Q but the symbols come from P .

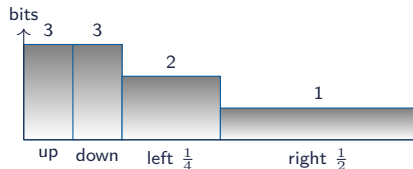
- Rover: $Q = (\frac{1}{2}, \frac{1}{4}, \frac{1}{8}, \frac{1}{8})$ is what the code was built for, $P = (\frac{1}{8}, \frac{1}{8}, \frac{1}{4}, \frac{1}{2})$ is the truth, and $H(P, Q) = 2.625$.

Cross-Entropy: Widths From P , Heights From Q

$H(P, Q) = 2.625$: widths P , heights $\log \frac{1}{Q}$



$H(P, P) = 1.75$: widths P , heights $\log \frac{1}{P}$



- $H(P, P) = H(P)$: the code built for the truth. The lower bound of Section 4 reads $H(P, Q) \geq H(P)$ for every Q (Gibbs' lemma with $a = Q$).
- $H(P, Q) \neq H(Q, P)$ in general: swapping truth and code swaps widths and heights, which changes the area.
- $H(P, Q) = \infty$ if $Q(x) = 0$ for some x with $P(x) > 0$: a code with no codeword for a symbol that does occur.

The same bars, re-weighted: the wide bar now carries the tall height, and the area grows from 1.75 to 2.625.

Cross-Entropy: Two Coins

A hedging code on a lopsided truth

$Q = (\frac{1}{2}, \frac{1}{2})$, $P = (0.9, 0.1)$. Lengths $\log 2 = 1$ and 1: $H(P, Q) = 0.9 \cdot 1 + 0.1 \cdot 1 = 1$ bit. The same for every P : equal lengths cost 1 bit whatever the truth. But $H(P) = h(0.1) = 0.469$, so 0.531 bits per symbol are wasted.

A confident code on a fair truth

$Q = (0.9, 0.1)$, $P = (\frac{1}{2}, \frac{1}{2})$. Lengths $\log \frac{1}{0.9} = 0.152$ and $\log \frac{1}{0.1} = 3.322$:

$$H(P, Q) = \frac{1}{2} \cdot 0.152 + \frac{1}{2} \cdot 3.322 = 1.737 \text{ bits,}$$

against $H(Q) = 0.469$, what the code was designed to achieve, and $H(P) = 1$, what the truth needs.

truth P	code Q	$H(P, Q)$	$H(P)$	waste $H(P, Q) - H(P)$
$(0.9, 0.1)$	$(\frac{1}{2}, \frac{1}{2})$	1	0.469	0.531
$(\frac{1}{2}, \frac{1}{2})$	$(0.9, 0.1)$	1.737	1	0.737

Hedging when the truth is lopsided wastes half a bit; being confidently wrong when the truth is fair wastes three quarters, because the rare-looking outcome costs 3.3 bits and comes half the time.

Exercise

1. The rover keeps the code 0, 10, 110, 111. Compute the cost per instruction if the truth becomes $(\frac{1}{4}, \frac{1}{4}, \frac{1}{4}, \frac{1}{4})$, and if it becomes $(\frac{1}{4}, \frac{1}{2}, \frac{1}{8}, \frac{1}{8})$.
2. Show that for Q uniform on n symbols, $H(P, Q) = \log n$ for every P . Conclude that the fixed-length code is the only code whose cost does not depend on the truth.
3. $P = (\frac{1}{2}, \frac{1}{2})$, $Q = (\frac{1}{4}, \frac{3}{4})$. Compute $H(P, Q)$ and $H(Q, P)$. Say in words which code is being used on which truth in each case.
4. Deduce $H(P, Q) \geq H(P)$, with equality iff $Q = P$, from Gibbs' lemma. Which choice of a is needed, and why does it satisfy $\sum a \leq 1$?
5. Every day a forecaster announces "rain with probability q "; the truth is rain with probability p , independently each day. Write her average cost in bits per day if her announcements are used as a code, and find the q that minimises it.

Cross-entropy prices a code, or a forecast, against a truth it was not built for.

Relative Entropy (D)

Definition

For pmfs P, Q on S , the **relative entropy** (Kullback–Leibler divergence) of P from Q is

$$D(P \parallel Q) = H(P, Q) - H(P) = \sum_{x \in S_P} P(x) \log \frac{P(x)}{Q(x)} \quad \text{bits, with } \log \frac{P(x)}{0} = \infty,$$

a function $D: \Delta_n \times \Delta_n \rightarrow [0, \infty]$; P is the truth, Q the code or model.

The waste: the extra bits per symbol paid for coding with Q when the truth is P , over the $H(P)$ that the truth needs.

The rover

$D(P \parallel Q) = 2.625 - 1.75 = 0.875$ bits per instruction wasted by keeping the old code. Bar by bar,
 $P(x) \left[\log \frac{1}{Q(x)} - \log \frac{1}{P(x)} \right]$:
 $\frac{1}{8}(1 - 3) + \frac{1}{8}(2 - 3) + \frac{1}{4}(3 - 2) + \frac{1}{2}(3 - 1) = -0.25 - 0.125 + 0.25 + 1 = 0.875$. Two bars got cheaper and two dearer; the dearer ones are the likely ones, so the total is positive.

- Two coins: $D\left((0.9, 0.1) \parallel \left(\frac{1}{2}, \frac{1}{2}\right)\right) = 0.531$ and $D\left(\left(\frac{1}{2}, \frac{1}{2}\right) \parallel (0.9, 0.1)\right) = 0.737$, the waste column of the previous table.

Gibbs' Inequality

Theorem (Gibbs' inequality)

$D(P \parallel Q) \geq 0$, with equality if and only if $P = Q$. Equivalently, $H(P, Q) \geq H(P)$, with equality iff $Q = P$.

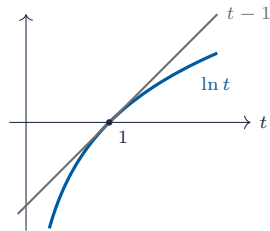
No code beats the code built for the truth, and no model predicts a source better on average than the source's own distribution.

Proof

If $Q(x) = 0 < P(x)$ for some x then $D = \infty > 0$. Otherwise, with $\ln t \leq t - 1$ at $t = Q(x)/P(x)$ (the lemma of Section 4 with $a = Q$):

$$\begin{aligned} -D(P \parallel Q) \ln 2 &= \sum_{x \in S_P} P(x) \ln \frac{Q(x)}{P(x)} \leq \sum_{x \in S_P} P(x) \left(\frac{Q(x)}{P(x)} - 1 \right) \\ &= \sum_{x \in S_P} Q(x) - 1 \leq 0, \end{aligned}$$

with equality iff $Q = P$ on S_P and $\sum_{S_P} Q = 1$, i.e. $Q = P$. $\square$



$\ln t \leq t - 1$, touching only at $t = 1$: the whole proof.

Consequences of Gibbs' Inequality

Corollary 1: the maximum of entropy

If X takes n values then $H(X) \leq \log n$, with equality iff X is uniform.

Proof

Take Q uniform, $Q(x) = \frac{1}{n}$: $0 \leq D(p_X \| Q) = \sum_x p_X(x) \log(n p_X(x)) = \log n - H(X)$, with equality iff $p_X = Q$. $\square$

Spreading probability evenly maximises the average surprise; the fixed-length code is optimal exactly for the uniform source.

Corollary 2: cross-entropy is minimised by the truth

For fixed P , the function $Q \mapsto H(P, Q)$ on Δ_n has its unique minimum at $Q = P$, where it equals $H(P)$. This is the whole reason cross-entropy can serve as a training loss (Section 6).

Corollary 3, for Section 8

Mutual information is non-negative: $I(X; Y) = D(p_{X,Y} \| p_X p_Y) \geq 0$, with equality iff $X \perp\!\!\!\perp Y$.

- Exercise 3 of the next battery does Corollary 1 by hand; Section 8 proves Corollary 3.

What Relative Entropy Is Not: A Distance

A distance d has $d(P, Q) = d(Q, P)$, $d(P, R) \leq d(P, Q) + d(Q, R)$, $d < \infty$, and $d(P, Q) = 0$ iff $P = Q$. Relative entropy has only the last.

property	fails at	numbers
symmetry	$P = (\frac{1}{2}, \frac{1}{2}), Q = (\frac{1}{4}, \frac{3}{4})$	$D(P\ Q) = 1.2075 - 1 = 0.2075$: the 2-bit codeword comes half the time instead of a quarter. $D(Q\ P) = 1 - 0.8113 = 0.1887$: 1 bit always, against the 0.811 needed.
triangle inequality	$P = (\frac{1}{2}, \frac{1}{2}), Q = (0.9, 0.1), R = (0.99, 0.01)$	$D(P\ Q) + D(Q\ R) = 0.737 + 0.208 = 0.945 < 2.329 = D(P\ R)$.
finiteness	$Q(x) = 0 < P(x)$	$D(P\ Q) = \infty$: the model said “never” and it happened; practical models never assign exactly 0.
$D = 0$ iff $P = Q$	holds (Gibbs)	D detects difference, measured in bits of waste for one direction of use.

Relative entropy is a divergence: it says how costly it is to mistake Q for P , and mistakes are not symmetric.

Exercise

1. Compute $D(P\|Q)$ and $D(Q\|P)$ for $P = (0.9, 0.1)$, $Q = (\frac{1}{2}, \frac{1}{2})$, once from $\sum P \log(P/Q)$ and once as $H(P, Q) - H(P)$. Check against the two-coins table.
2. Repeat for $P = (\frac{1}{2}, \frac{1}{2})$, $Q = (\frac{1}{4}, \frac{3}{4})$, and explain in one sentence why $D(P\|Q) > D(Q\|P)$ here.
3. Show that $D(P\|U) = \log n - H(P)$ for U uniform, and read off the maximum of H on Δ_n and where it is attained.
4. The rover's engineers propose a compromise code built for $Q = (\frac{1}{4}, \frac{1}{4}, \frac{1}{4}, \frac{1}{4})$, to be used whether the truth is $(\frac{1}{2}, \frac{1}{4}, \frac{1}{8}, \frac{1}{8})$ or $(\frac{1}{8}, \frac{1}{8}, \frac{1}{4}, \frac{1}{2})$. Compute the waste in each case and compare with keeping the old code.
5. Show that $D(P\|Q)$ is unchanged by relabelling the symbols but that, unlike a distance, $D(P\|Q) \leq \varepsilon$ does not imply $D(Q\|P) \leq \varepsilon$: give P, Q with $D(P\|Q) < 0.1$ and $D(Q\|P) > 1$.

Relative entropy is the number to compute when deciding which of several codes, or models, to install.

Why the Loss Is a Logarithm

A classifier outputs a pmf $q \in \Delta_n$ over $n \geq 3$ classes; the true class is drawn from $p \in \Delta_n$. When class y occurs, charge $f(q_y)$, with $f: (0, 1] \rightarrow \mathbb{R}$ decreasing and differentiable on $(0, 1)$.

Theorem

Suppose f is **honest**: for every truth p , the expected penalty $\sum_y p_y f(q_y)$ over $q \in \Delta_n$ is smallest at $q = p$ and nowhere else. Then $f(t) = -c \log t + d$ on $(0, 1)$ with $c > 0$; normalised, $f(t) = -\log t$, the information of the outcome under the model.

Proof

Take p with every $p_y > 0$. Then $q = p$ is a minimiser in the interior of Δ_n , so Lagrange applies there: $p_y f'(p_y) = \lambda$ for all y , i.e. $t f'(t)$ takes the same value at every coordinate $t = p_y$. For $s, t > 0$ with $s + t < 1$ use $p = (s, t, \frac{1-s-t}{n-2}, \dots, \frac{1-s-t}{n-2})$ (possible as $n \geq 3$): $s f'(s) = t f'(t)$. For $s + t \geq 1$ compare both with a small u . So $t f'(t) = -c$ is constant on $(0, 1)$, $f(t) = -c \ln t + d$, and decreasing means $c > 0$. $\square$

A model that is to report what it believes must be charged the logarithm of the probability it gave to what happened; nothing else works.

What Goes Wrong With Any Other Penalty

Linear penalty $f(t) = 1 - t$

Expected penalty $1 - \sum_y p_y q_y$, smallest when all of q sits on the most probable class: for $p = (0.5, 0.3, 0.2)$ the minimiser is $q = (1, 0, 0)$, penalty 0.5, while honesty costs 0.62. The model learns to be overconfident.

Squared penalty on the realised probability, $f(t) = (1 - t)^2$

Lagrange: $-2p_y(1 - q_y) = \lambda$, so $q_y = 1 + \lambda/(2p_y)$. For $p = (0.5, 0.3, 0.2)$: $q = (0.613, 0.355, 0.032) \neq p$. Again the likely is exaggerated and the unlikely buried. (The Brier score penalises the whole vector q , not just q_y , and is honest.)

Logarithmic penalty $f(t) = -\log t$

Expected penalty $\sum_y p_y \log \frac{1}{q_y} = H(p, q) = H(p) + D(p||q)$, minimised at $q = p$ by Gibbs, and only there. For $p = (0.5, 0.3, 0.2)$ the minimum is $H(p) = 1.485$ bits.

Only the logarithm makes honesty optimal; every other local penalty rewards some distortion of the truth.

Cross-Entropy Loss

Definition (cross-entropy loss)

Data $(x_1, y_1), \dots, (x_N, y_N)$, labels among n classes; a model with parameters $\theta \in \Theta \subseteq \mathbb{R}^d$ outputs a pmf $q_\theta(\cdot | x) \in \Delta_n$ over the classes for each input x . Its loss is

$$\mathcal{L}(\theta) = \frac{1}{N} \sum_{i=1}^N \log \frac{1}{q_\theta(y_i | x_i)} \quad \text{bits per example.}$$

Each example is charged the information of its true label under the model: the codeword length the model would have used for it.

What minimising it does

For an input x whose true label has pmf $p(\cdot | x)$, the loss averages to $H(p(\cdot | x), q_\theta(\cdot | x)) = H(p(\cdot | x)) + D(p(\cdot | x) \parallel q_\theta(\cdot | x))$. The first term does not depend on θ . Minimising the loss is minimising D : making the model match the truth, and by Gibbs nothing else will do it.

- The loss cannot go below $H(p)$: a curve flat at 0.5 bits is not stuck if the labels carry 0.5 bits of genuine uncertainty per example.

Distillation: Training on a Whole Distribution

A large *teacher* model gives, for each input x , a whole pmf $p(\cdot | x)$ over classes. A small *student* q_θ is trained to minimise

$$H(p(\cdot | x), q_\theta(\cdot | x)) = \sum_y p(y | x) \log \frac{1}{q_\theta(y | x)} \quad \text{averaged over inputs.}$$

The same cross-entropy loss, with the teacher's soft probabilities in place of a single correct label.

- A one-hot label has entropy 0: it says which class, and nothing about how the alternatives rank. The teacher's pmf has $H(p) > 0$ bits, and every one of them is information about the input that the student can learn from.
- The loss floor is $H(p)$, not 0; the student's distance from the teacher is $D(p||q_\theta)$, the part it can actually reduce.

Numbers

Teacher $p = (0.7, 0.2, 0.1)$, student $q = (0.8, 0.1, 0.1)$:

$H(p, q) = 0.7 \log \frac{1}{0.8} + 0.2 \log \frac{1}{0.1} + 0.1 \log \frac{1}{0.1} = 1.222$ bits $= H(p) + D(p||q) = 1.157 + 0.065$. The student is 0.065 bits from the teacher; the remaining 1.157 is the teacher's own uncertainty.

Prediction and Compression Are the Same Thing

Theorem (arithmetic coding; statement only)

Let a model assign to a text $x_1 \dots x_n$ the conditional probabilities $q(x_i \mid x_1 \dots x_{i-1}) > 0$. There is a uniquely decodable encoding of the text in fewer than

$$\sum_{i=1}^n \log \frac{1}{q(x_i \mid x_1 \dots x_{i-1})} + 2 \text{ bits},$$

decodable by anyone who has the same model.

A predictor is a compressor: each symbol costs the information the model assigned to it, and the total is the model's cross-entropy loss on the text, plus two bits.

- The training loss of a language model, in bits per token, is the size to which it compresses its training text, per token. Lower loss, smaller file, better predictions: one number.
- Conversely a compressor defines a predictor: a symbol coded with ℓ bits has been assigned probability $2^{-\ell}$. No model gets below the source's entropy rate (Section 9) on average, by Gibbs.
- The rover: the model “up $\frac{1}{2}$, down $\frac{1}{4}$, left, right $\frac{1}{8}$ ” compresses instructions to 1.75 bits each; the reversed model costs 2.625 on the same stream.

Exercise

1. Two classes, truth $p = 0.9$. Compute the expected log loss of models reporting $q = 0.5, 0.9, 0.99$. Which is honest, and why is 0.99 punished although it is “more confident in the right answer”?
2. A forecaster believes rain has probability 0.3 and is scored by log loss. Show her expected score is minimised by announcing 0.3, and compute the expected cost of hedging with 0.5.
3. A model's loss has flattened at 0.5 bits per example. Describe two situations that produce this, one in which the model should be improved and one in which it cannot be.
4. For $p = (0.5, 0.3, 0.2)$, verify by Lagrange multipliers that the penalty $f(t) = (1 - t)^2$ on the realised probability is minimised at $q = (0.613, 0.355, 0.032)$, and compute the penalty there and at $q = p$.
5. Explain why the theorem needs $n \geq 3$: for two classes, show that $f(t) = (1 - t)^2$ does satisfy the honesty requirement.

The log loss is the only penalty on the realised probability that rewards a model for reporting what it believes.

Motivation: Which Wordle Opener?

Wordle: a five-letter answer is drawn from a list of 2315 words. Each guess returns five colours, ■ right letter in the right place, ■ right letter in the wrong place, ■ letter absent: one of $3^5 = 243$ patterns.

Head-on

Before the first guess the answer is uniform on 2315 words: $\log 2315 = 11.18$ bits of uncertainty. A guess “is worth” as much as it reduces this. But the reduction depends on which pattern comes back, and there are 243 of them.

Questions we cannot yet answer

1. Is weary a good first guess? Is slate better? Is anything better than that?
2. Can any strategy average fewer than 3 guesses?
3. Why did the “best opener” change when people optimised for the actual score instead of information?

The tool is the entropy of the pattern distribution of a guess, which is the expected information the guess delivers, and which Section 8 will identify as a mutual information.

- Source of the real-game numbers below: 3Blue1Brown, “Solving Wordle using information theory” (2022).

A Guess Is a Partition of the Remaining Answers

Definition

Let W be the set of remaining possible answers, $|W| = N$, and let the answer $A: \Omega \rightarrow W$ be uniform on W : $\mathbb{P}(A = w) = \frac{1}{N}$. Fix a guess g . The **pattern function** of g is $\pi_g: W \rightarrow \{\text{grey}, \text{orange}, \text{green}\}^5$, sending each possible answer w to the colours g would receive if w were the answer. The **pattern** is the random variable $\Pi = \pi_g(A): \Omega \rightarrow \{\text{grey}, \text{orange}, \text{green}\}^5$, with pmf

$$p_{\Pi}(\pi) = \mathbb{P}(\Pi = \pi) = \frac{|\{w \in W : \pi_g(w) = \pi\}|}{N} = \frac{|\text{bucket of } \pi|}{N}.$$

A guess sorts the remaining answers into buckets, one per pattern; the pattern you get back tells you which bucket the answer is in, and nothing more.

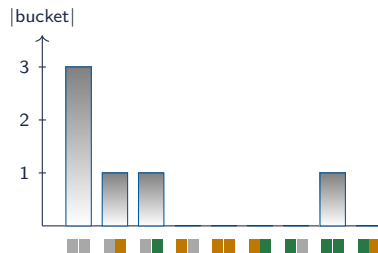
- Seeing pattern π shrinks the candidate list from N to $|\text{bucket of } \pi| = N p_{\Pi}(\pi)$. The information of the pattern is $I_{\Pi}(\pi) = \log \frac{1}{p_{\Pi}(\pi)} = \log \frac{N}{|\text{bucket}|}$: the number of halvings of the list.
- The guess is fully described by its pattern distribution p_{Π} : a bar chart with 243 bars. Two guesses with the same bar chart are equally informative, whatever the letters.
- A pattern with a bucket of size 1 carries $\log N$ bits and ends the game next turn; the all-grey pattern usually has a huge bucket and carries little.

Toy Wordle: Two-Letter Words, Done by Hand

$W = \{AM, IN, NO, OF, ON, TO\}$, $N = 6$, two-letter words,
 $3^2 = 9$ possible patterns. Guess $g = IN$; for each possible
 answer, the pattern:

answer w	$\pi_{IN}(w)$	why
AM	■	no I, no N
IN	■	the answer
NO	■	N, wrong place
OF	■	
ON	■	N in place
TO	■	

Buckets: ■ $\{AM, OF, TO\}$, and ■, ■, ■ with one word
 each. So $p_{\Pi} = (\frac{3}{6}, \frac{1}{6}, \frac{1}{6}, \frac{1}{6})$ on these four patterns, 0 on the
 other five.



The bar chart of the guess IN: all nine
 patterns, four of them possible. Bucket of
 3: $I = \log \frac{6}{3} = 1$ bit. Buckets of 1:
 $I = \log 6 = 2.585$ bits.

The Entropy of a Guess Is Its Expected Information

Definition

The **entropy of the guess** g is the entropy of its pattern, $H(\Pi) = \sum_{\pi} p_{\Pi}(\pi) \log \frac{1}{p_{\Pi}(\pi)} = \mathbb{E}[I_{\Pi}(\Pi)]$, the expected number of halvings of the list.

Average the information of each pattern you might get back: what the guess is worth before you make it.

Toy Wordle

guess	bucket sizes	$H(\Pi)$
NO or ON	1, 1, 1, 1, 1, 1	$\log 6 = 2.585$ bits, the maximum: each pattern names the answer
OF or TO	2, 2, 1, 1	$2 \cdot \frac{2}{6} \cdot 1.585 + 2 \cdot \frac{1}{6} \cdot 2.585 = 1.918$ bits
IN	3, 1, 1, 1	$\frac{3}{6} \cdot 1 + 3 \cdot \frac{1}{6} \cdot 2.585 = 1.792$ bits
AM	5, 1	$\frac{5}{6} \cdot 0.263 + \frac{1}{6} \cdot 2.585 = 0.650$ bits: wasted five times in six

Real Wordle, $N = 2315$ (3Blue1Brown)

weary 4.90 bits; slate 5.87; soare 5.89, the best by this measure, leaving $2315/2^{5.89} \approx 39$ candidates on average. The ceiling $\log 243 = 7.92$ (one word per pattern) is unreachable since $2315 > 243$.

Back to the Openers: Two Guesses Almost Never Finish

Let $G: \Omega \rightarrow \{1, 2, \dots\}$ be the number of guesses a strategy takes to hit the answer; the score of the strategy is $\mathbb{E}[G]$.

Claim

For every strategy, $\mathbb{P}(G \leq 2) \leq \frac{243}{2315} = 0.105$, and hence $\mathbb{E}[G] \geq 3 - 2\mathbb{P}(G = 1) - \mathbb{P}(G = 2) \geq 3 - \frac{244}{2315} = 2.89$.

Proof

$G = 1$ for at most one answer, the first guess itself. $G = 2$ requires the second guess to be the answer; the second guess is a function of the first pattern, which takes at most 242 non-winning values, so at most 242 answers have $G = 2$. Then $\mathbb{E}[G] \geq 1 \cdot \mathbb{P}(G = 1) + 2\mathbb{P}(G = 2) + 3\mathbb{P}(G \geq 3)$. $\square$

A guess can only single out one answer per pattern it can receive, so two guesses cover at most 243 of 2315 answers.

- In bits: $\log 2315 = 11.18$; the best opening pair extracts 10.04 bits on average (3Blue1Brown), leaving 1.14 bits, i.e. $2^{1.14} \approx 2.2$ candidates for guess three. The best strategy known averages 3.42 guesses.
- Toy Wordle: open NO, $\mathbb{E}[G] = \frac{1}{6} \cdot 1 + \frac{5}{6} \cdot 2 = 1.83$; open IN then OF on , $\mathbb{E}[G] = \frac{1+2+2+2+3+3}{6} = 2.17$. Here the entropy ranking and the score ranking agree.

Back to the Openers: Entropy Is a Proxy for the Score

Bits versus guesses

$H(\Pi)$ is what the guess tells you; $\mathbb{E}[G]$ counts turns. They differ because

- a guess $g \in W$ can end the game now ($\mathbb{P}(G = \text{this turn}) = \frac{1}{N}$); a guess $g \notin W$ cannot, however large $H(\Pi)$;
- after the last turn, bits are worthless: only whether the list has size 1 matters.

Maximising $H(\Pi)$ each turn is a proxy for minimising $\mathbb{E}[G]$.

Entropy measures information gained per guess; the game counts guesses; the two agree closely, and not exactly.

- Section 8 identifies $H(\Pi)$ as the mutual information $I(A; \Pi)$ between the answer and the pattern, which is what “expected information of a guess” means precisely.

opener	$H(\Pi)$	$\mathbb{E}[G]$, best play
soare	5.89 (max)	
slate	5.87	
salet	< 5.89	3.42 (min)
crane	< 5.89	3.43
weary	4.90	

Scores by full game-tree search (3Blue1Brown, 2022): the best opener by score is not the best by entropy.

Exercise

1. On the toy list, write out the pattern table for the opener OF and compute its bucket sizes and $H(\Pi)$. Do the same for AM.
2. After IN returns ■■■, $W = \{\text{AM}, \text{OF}, \text{TO}\}$. Which second guesses from the list identify the answer for certain? Compute their entropies on the reduced list.
3. Show that $H(\Pi) \leq \log N$ for every guess, with equality iff every bucket has size 1; and that $H(\Pi) \leq \log 243$ in real Wordle. Which bound bites at the start, and which near the end?
4. A guess $g \in W$ has a bucket $\{g\}$ of size 1 for the pattern ■■■■■. Show that removing g from W and re-guessing changes $H(\Pi)$, and explain why a guess outside W can have higher entropy than any guess inside W .
5. Explain, without computing, why the guess maximising $H(\Pi)$ need not minimise the expected number of guesses. What extra information about the buckets would you need to compute the expected number of guesses exactly?

The entropy of a guess is what it tells you, computable before you make it, not what it wins you.

The Two-Dice Table, and Joint Entropy

Two fair dice, $\mathbb{P}(\{\omega\}) = \frac{1}{36}$ on $\Omega = \{1, \dots, 6\}^2$. Y the first die (row), Z the second (column), $X = Y + Z$ (the cell).
The rows are the events $\{Y = y\}$.

Definition

The **joint entropy** of X and Y is the entropy of the pair, i.e. of $p_{X,Y}: S_X \times S_Y \rightarrow [0, 1]$,
 $p_{X,Y}(x, y) = \mathbb{P}(X = x, Y = y)$:

$$H(X, Y) = \sum_{x,y} p_{X,Y}(x, y) \log \frac{1}{p_{X,Y}(x, y)}.$$

The bits needed to report both values at once.

Two dice

$p_{X,Y}(x, y) = \frac{1}{36}$ if $x - y \in \{1, \dots, 6\}$, else 0: one cell per pair that occurs. So (X, Y) is uniform on 36 pairs and $H(X, Y) = \log 36 = 5.170$ bits: the sum and the first die together name the cell.

$y \setminus z$	1	2	3	4	5	6
1	2	3	4	5	6	7
2	3	4	5	6	7	8
3	4	5	6	7	8	9
4	5	6	7	8	9	10
5	6	7	8	9	10	11
6	7	8	9	10	11	12

Rows y , columns z , cell $x = y + z$. Row 4 is the event $\{Y = 4\}$.

$H(X \mid Y = y)$: The Entropy of a Row

Definition

For $y \in S_Y$, let $p_{X|Y}(\cdot \mid y)$ be the conditional pmf of X given $Y = y$, the row's distribution: $p_{X|Y}(x \mid y) = p_{X,Y}(x, y)/p_Y(y)$. The **conditional entropy of X given $Y = y$** is the function

$$S_Y \rightarrow [0, \infty), \quad y \mapsto H(X \mid Y = y) = \sum_x p_{X|Y}(x \mid y) \log \frac{1}{p_{X|Y}(x \mid y)}.$$

Restrict to row y and take the entropy of what you see there: a number for each row.

Two dice, X the sum

Row y contains the six distinct sums $y + 1, \dots, y + 6$, each with conditional probability $\frac{1}{6}$. So

$H(X \mid Y = y) = \log 6 = 2.585$ bits for every y .

- The other way round: $\{X = x\}$ is an anti-diagonal of n_x cells with Y uniform on it, so $H(Y \mid X = x) = \log n_x$, which runs 0, 1, 1.585, 2, 2.322, 2.585 and back down as x goes from 2 to 12.

$y \setminus z$	1	2	3	4	5	6
1	2	3	4	5	6	7
2	3	4	5	6	7	8
3	4	5	6	7	8	9
4	5	6	7	8	9	10
5	6	7	8	9	10	11
6	7	8	9	10	11	12

$\{X = 7\}$: six cells, $H(Y \mid X = 7) = \log 6$.

$H(X | Y)$: The Average Over the Rows

Definition

The **conditional entropy of X given Y** is the row entropy averaged over the rows with their probabilities,

$$H(X | Y) = \sum_{y \in S_Y} p_Y(y) H(X | Y = y) = \sum_{x,y} p_{X,Y}(x,y) \log \frac{1}{p_{X|Y}(x | y)} \quad \text{a number.}$$

The uncertainty left in X once Y is known, averaged over what Y turns out to be: total expectation applied to row entropies.

- **What it is not.** Unlike $\mathbb{E}[X | Y]$, a random variable, $H(X | Y)$ is a *number*: the row entropies have been averaged. The notation is parallel; the type is not.

Two dice

$H(X | Y) = \sum_y \frac{1}{6} \log 6 = 2.585$ bits: every row of sums is uniform on six values.

$H(Y | X) = \sum_x \frac{n_x}{36} \log n_x = 2 \left[\frac{2}{36} \cdot 1 + \frac{3}{36} \cdot 1.585 + \frac{4}{36} \cdot 2 + \frac{5}{36} \cdot 2.322 \right] + \frac{6}{36} \cdot 2.585 = 1.896$ bits: the sum leaves less than a die's worth of doubt about the first die.

The Chain Rule

Theorem (chain rule)

$$H(X, Y) = H(Y) + H(X | Y) = H(X) + H(Y | X).$$

Proof

$p_{X,Y}(x, y) = p_Y(y) p_{X|Y}(x | y)$, so $\log \frac{1}{p_{X,Y}(x,y)} = \log \frac{1}{p_Y(y)} + \log \frac{1}{p_{X|Y}(x|y)}$. Multiply by $p_{X,Y}(x, y)$ and sum over (x, y) . The first term: $\sum_y \log \frac{1}{p_Y(y)} \sum_x p_{X,Y}(x, y) = \sum_y p_Y(y) \log \frac{1}{p_Y(y)} = H(Y)$. The second term is $H(X | Y)$ by definition. Swap the roles of X and Y for the other form. $\square$

To report the cell, report the row, then the position within the row: the bits add.

Two dice, both ways

$$\log 36 = 5.170 = \underbrace{2.585}_{H(Y)} + \underbrace{2.585}_{H(X|Y)} = \underbrace{3.274}_{H(X)} + \underbrace{1.896}_{H(Y|X)}.$$

- Extremes: $X \perp\!\!\!\perp Y$ gives $H(X | Y) = H(X)$ and $H(X, Y) = H(X) + H(Y)$ (Section 4); X a function of Y gives $H(X | Y) = 0$ and $H(X, Y) = H(Y)$.

Exercise

1. Two dice. Compute $H(Z | Y = y)$ for each row and hence $H(Z | Y)$. Check $H(Y, Z) = H(Y) + H(Z | Y)$. What does $H(Z | Y) = H(Z)$ say about the rows?
2. Two dice. Tabulate $x \mapsto H(Y | X = x)$ for $x = 2, \dots, 12$ and verify $H(Y | X) = 1.896$ by averaging with p_X .
3. Let $M = \max(Y, Z)$. Compute $H(M | Y = y)$ for each y (which cells of row y have which maximum?) and hence $H(M | Y)$; then $H(M)$ from its pmf $(1, 3, 5, 7, 9, 11)/36$; then $H(Y | M)$ by the chain rule.
4. Prove $H(g(X)) \leq H(X)$ for any function g , by writing the chain rule for the pair $(X, g(X))$ both ways and using $H(g(X) | X) = 0$. When is it an equality?
5. A bit X uniform on $\{0, 1\}$ is sent down a wire that flips it with probability ε , independently, giving Y . Draw the 2×2 table of $p_{X,Y}$ (rows $\{X = 0\}, \{X = 1\}$). Compute $H(Y | X = x)$, $H(Y | X)$, $H(X, Y)$ and $H(Y)$.

Conditional entropy is row-then-average on the table, and the chain rule is how the bits of a pair are split.

Mutual Information (I)

Definition

The **mutual information** of X and Y is the uncertainty in X removed by learning Y :

$$I(X; Y) = H(X) - H(X | Y) \quad \text{bits, a number.}$$

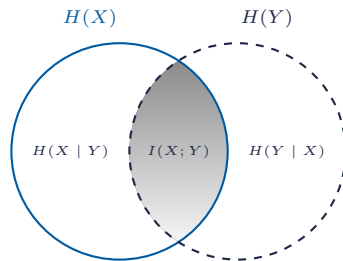
How many of X 's bits Y carries: what you knew about the cell's number before seeing the row, minus what is left to learn once you have.

Two dice

$I(X; Y) = H(X) - H(X | Y) = 3.274 - 2.585 = 0.689$ bits: the first die tells you about two thirds of a bit about the sum. The other way:

$I(Y; X) = H(Y) - H(Y | X) = 2.585 - 1.896 = 0.689$. The same number, and the next slide says why.

- Two dice in the picture: $H(X, Y) = 5.170$, split as $2.585 + 0.689 + 1.896$.



The whole picture is $H(X, Y)$. Left circle $H(X)$, right circle $H(Y)$, overlap $I(X; Y)$.

Mutual Information: Properties

Theorem

1. $I(X; Y) = H(X) + H(Y) - H(X, Y) = I(Y; X)$.
2. $I(X; Y) = D(p_{X,Y} \parallel p_X p_Y) = \sum_{x,y} p_{X,Y}(x, y) \log \frac{p_{X,Y}(x, y)}{p_X(x) p_Y(y)} \geq 0$, with equality iff $X \perp\!\!\!\perp Y$.
3. $H(X | Y) \leq H(X)$, with equality iff $X \perp\!\!\!\perp Y$; and $H(X, Y) \leq H(X) + H(Y)$.
4. $I(X; X) = H(X)$, and $I(X; Y) \leq \min(H(X), H(Y))$.

Proof

(1) Chain rule: $H(X | Y) = H(X, Y) - H(Y)$; substitute; the result is symmetric. (2) $H(X) - H(X | Y) = \sum_{x,y} p_{X,Y}(x, y) \left[\log \frac{1}{p_X(x)} - \log \frac{p_Y(y)}{p_{X,Y}(x, y)} \right]$, the displayed sum, which is D of the joint pmf from the product pmf; Gibbs gives ≥ 0 , with equality iff $p_{X,Y} = p_X p_Y$: independence. (3) Rearrange (2) and (1). (4) $H(X | X) = 0$; $H(X | Y) \geq 0$ gives $I \leq H(X)$, and symmetrically $I \leq H(Y)$. □

Knowing Y never increases the uncertainty about X on average, though a single row can ($M = \max(Y, Z)$ of the last battery: $H(M) = 2.320$ but $H(M | Y = 1) = \log 6 = 2.585$), and it leaves it unchanged only when every row looks like the whole table.

Example: The Binary Symmetric Channel

A bit X , uniform on $\{0, 1\}$, is sent down a wire that flips it with probability ε , independently of X ; Y is what arrives.

How many bits about X does one received bit carry?

- Head-on: “ $1 - \varepsilon$ of a bit” is the natural guess. It is wrong.
- Rows $\{X = 0\}$ and $\{X = 1\}$: each row's distribution for Y is $(1 - \varepsilon, \varepsilon)$ or $(\varepsilon, 1 - \varepsilon)$, entropy $h(\varepsilon)$. So $H(Y | X = x) = h(\varepsilon)$ for both rows and $H(Y | X) = h(\varepsilon)$.
- Y is uniform by symmetry: $H(Y) = 1$.

$I(X; Y) = H(Y) - H(Y | X) = 1 - h(\varepsilon)$ bits per received bit.

$p_{X,Y}$	$Y = 0$	$Y = 1$
$X = 0$	$\frac{1-\varepsilon}{2}$	$\frac{\varepsilon}{2}$
$X = 1$	$\frac{\varepsilon}{2}$	$\frac{1-\varepsilon}{2}$
ε	$h(\varepsilon)$	$I(X; Y)$
0	0	1
0.01	0.081	0.919
0.1	0.469	0.531
0.25	0.811	0.189
0.5	1	0

A wire that flips one bit in ten delivers about half a bit of information per bit sent, not nine tenths: the receiver does not know which bits were flipped.

- $1 - h(\varepsilon)$ is the capacity of this channel (Shannon's channel coding theorem, not proved here): the rate at which reliable communication is possible over it.

Back to Wordle: The Expected Information of a Guess Is $I(A; \Pi)$

Answer A uniform on W , $|W| = N$; guess g ; pattern $\Pi = \pi_g(A)$. Rows $\{A = w\}$, one per answer; the pattern is constant on each.

Theorem

$I(A; \Pi) = H(\Pi)$: the entropy of the guess (Section 7) is the mutual information between answer and pattern.

Proof

Π is a function of A : row $\{A = w\}$ has the single pattern $\pi_g(w)$, so $H(\Pi \mid A = w) = 0$ for every w , $H(\Pi \mid A) = 0$, and $I(A; \Pi) = H(\Pi) - 0$. By symmetry also

$I(A; \Pi) = H(A) - H(A \mid \Pi) = \log N - \sum_{\pi} p_{\Pi}(\pi) \log |\text{bucket of } \pi|$: the expected shrinkage of the list, in bits. □

The pattern has no randomness of its own, so what it tells you about the answer is all it has.

Toy Wordle, guess IN

$H(A) = \log 6 = 2.585$. With probability $\frac{1}{2}$ the bucket has size 3 (entropy $\log 3 = 1.585$), else size 1 (entropy 0): $H(A \mid \Pi) = 0.792$ and $I(A; \Pi) = 2.585 - 0.792 = 1.792 = H(\Pi)$, as in Section 7.

Exercise

1. Two dice. Compute $I(Y; Z)$, $I(X; Z)$ and $I(X; (Y, Z))$, each in one line from quantities already known.
2. The wire with $\varepsilon = 0.1$ is used twice on independent uniform bits. How many bits of information arrive in total? What if the same bit is sent twice (harder: find $H(X | Y_1, Y_2)$ by listing the four rows $\{(Y_1, Y_2) = (y_1, y_2)\}$)?
3. Show $I(X; Y) = H(Y)$ iff Y is a function of X , and $I(X; Y) = H(X) = H(Y)$ iff each is a function of the other.
4. X a fair die, N an independent fair coin in $\{0, 1\}$, $Y = X + N$. Compute $H(Y | X)$, $H(Y)$ and $I(X; Y)$. Why is $I(X; Y) < H(X)$ here although Y is “ X plus a little noise”?
5. In real Wordle, prove $I(A; \Pi) \leq \log 243$ for every guess, and $I(A; \Pi) \leq \log N$. Which of the two bounds explains why no opener can reach 11.18 bits?

Mutual information is the number of bits about one variable that actually arrive through another.

Motivation: Why Thirty Letters Break a Substitution Cipher

A **substitution cipher** replaces each letter by another under a fixed bijection k of the alphabet: $26! \approx 4 \cdot 10^{26}$ keys, $\log 26! = 88.4$ bits. Trying them all is hopeless. Yet a ciphertext of twenty-five to thirty letters usually has a unique sensible decryption, and a longer one falls to frequency counts in minutes.

Questions we cannot yet answer

1. How many bits does one letter of English carry? The head-on answer $\log 27 = 4.75$ (26 letters and a space) is too big by a factor of four or five.
2. How does a model of text turn that number into a decryption, a compression ratio, or a language-model score?
3. Why thirty letters, and not three or three hundred?

The tools are a Markov model of letters, its entropy rate, and the cross-entropy of a model on real text.

- Shannon, “Prediction and Entropy of Printed English” (1951), is the source of every English figure quoted below.

Three Models of Text

Text is a sequence of random variables $X_1, X_2, \dots : \Omega \rightarrow S$, $|S| = 27$ (letters and space). A model specifies the joint pmf of every stretch $X_1 \dots X_n$.

Definition

1. **Monographic (order 1).** A pmf $p: S \rightarrow [0, 1]$ of single letters; the X_i are i.i.d. with pmf p :
 $\mathbb{P}(X_1 = x_1, \dots, X_n = x_n) = \prod_i p(x_i)$.
2. **Digraphic (order 2).** A pmf $p_2: S \times S \rightarrow [0, 1]$ of adjacent pairs,
 $p_2(x, x') = \mathbb{P}(X_i = x, X_{i+1} = x')$, the same for every i ; its marginal is p .
3. **Markov. A transition matrix** $Q: S \times S \rightarrow [0, 1]$ with $\sum_{x'} Q(x, x') = 1$ for every x , where
 $Q(x, x') = \mathbb{P}(X_{i+1} = x' \mid X_i = x) = p_2(x, x')/p(x)$ for $p(x) > 0$ (any row for $p(x) = 0$), and

$$\mathbb{P}(X_1 = x_1, \dots, X_n = x_n) = p(x_1) Q(x_1, x_2) Q(x_2, x_3) \cdots Q(x_{n-1}, x_n).$$

The first model knows E is common and Z rare; the second knows TH and HE are common and Q is followed by U; the third turns the pair table into a rule for writing text one letter at a time.

- Q is a table of the two-dice kind: row x is the event $\{X_i = x\}$, and the numbers along row x are the conditional pmf of the next letter. Conditioning on the previous letter is restricting to a row.

Entropy Rate of a Markov Model

Definition

A pmf π on S is **stationary** for Q if $\sum_x \pi(x)Q(x, x') = \pi(x')$ for all x' : letter frequencies do not drift. For the Markov model started from π , the **entropy rate** is

$$H(X_{i+1} \mid X_i) = \sum_{x \in S} \pi(x) H(X_{i+1} \mid X_i = x) = \sum_{x \in S} \pi(x) H(Q(x, \cdot)) \quad \text{bits per letter,}$$

the entropy of row x of Q , averaged over rows with the row probabilities $\pi(x)$.

The bits needed for the next letter when the previous one is known: row-then-average, as for $H(X \mid Y)$, applied to the transition table.

- **What it is not.** Not $H(\pi)$, the entropy of a letter seen alone; and not the entropy of any one row. It is the average row entropy, and it is $H(X_{i+1} \mid X_i)$ for every i because π is stationary.
- n letters have entropy $H(X_1, \dots, X_n) = H(\pi) + (n-1)H(X_{i+1} \mid X_i)$, by the chain rule for n variables (exercise 4 below, by induction) and the Markov property $H(X_{k+1} \mid X_1, \dots, X_k) = H(X_{k+1} \mid X_k)$: each letter depends on the past only through the last letter (stated without proof). Per letter this tends to the entropy rate, the floor for compressing the model's text (Section 4 applied to blocks).

Context Never Hurts: Entropy Rate at Most Monographic Entropy

Theorem

$H(X_{i+1} | X_i) \leq H(\pi)$, with equality if and only if every row of Q equals π , i.e. consecutive letters are independent.

Proof

“Conditioning reduces entropy” (Section 8, property 3) with $X = X_{i+1}$ and $Y = X_i$, both of which have pmf π by stationarity: $H(X_{i+1} | X_i) \leq H(X_{i+1}) = H(\pi)$, with equality iff $X_{i+1} \perp\!\!\!\perp X_i$, i.e. $Q(x, \cdot) = \pi$ for every x . □

A model that looks at the previous letter can only lower the bits per letter, and it lowers them exactly when the previous letter carries information about the next.

- The saving $H(\pi) - H(X_{i+1} | X_i) = I(X_i; X_{i+1})$ is the mutual information between consecutive letters.
- Same argument with longer contexts: $H(X_{i+1} | X_i, X_{i-1}) \leq H(X_{i+1} | X_i)$, and so on. Shannon’s table on the slide after next is this chain of inequalities, measured on English.

Worked Example: Vowels and Consonants

$S = \{V, C\}$. After a vowel the next letter is a vowel with probability 0.2; after a consonant, 0.5.

Stationary pmf

Balance the flows $V \rightarrow C$ and $C \rightarrow V$:

$$\pi(V) \cdot 0.8 = \pi(C) \cdot 0.5, \quad \pi(V) + \pi(C) = 1:$$

$$\pi = \left(\frac{5}{13}, \frac{8}{13}\right) = (0.385, 0.615).$$

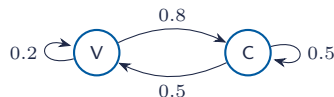
Entropy rate versus monographic entropy

$$\begin{aligned} H(X_{i+1} | X_i) &= \frac{5}{13} h(0.2) + \frac{8}{13} h(0.5) = \frac{5}{13}(0.722) + \frac{8}{13}(1) \\ &= 0.893 \text{ bits}, \quad H(\pi) = h\left(\frac{5}{13}\right) = 0.961 \text{ bits.} \end{aligned}$$

Knowing the previous letter saves 0.07 bits per letter here; in English the saving from context is several bits.

- Digraph check: Markov $\mathbb{P}(VV) = \pi(V)Q(V, V) = \frac{1}{13} = 0.077$; monographic $\pi(V)^2 = 0.148$, twice too often.

Q	V	C
V	0.2	0.8
C	0.5	0.5



Row entropies: $h(0.2) = 0.722$ and $h(0.5) = 1$.

Exercise

1. Suppose every row of Q is the same pmf π . Show that π is stationary, that the entropy rate equals $H(\pi)$, and that consecutive letters are independent. Which of the three models of text is this?
2. In the Markov model, write $\mathbb{P}(X_i = Q, X_{i+1} = U)$ in terms of p and Q ; do the same in the monographic model. Which is larger for English, and by roughly what factor if $Q(Q, U) \approx 1$ and $p(U) \approx 0.03$?
3. Change the vowel/consonant rule so that after a consonant the next letter is a vowel with probability 0.9. Recompute π and the entropy rate. Is this text more or less predictable than before, and than its own monographic model?
4. Prove the chain rule for n variables, $H(X_1, \dots, X_n) = \sum_{k=1}^n H(X_k \mid X_1, \dots, X_{k-1})$, by induction from the two-variable rule (treat $(X_1, \dots, X_{k-1})$ as one random variable). With the Markov property, deduce $H(X_1, \dots, X_n) = H(\pi) + (n-1)H(X_{i+1} \mid X_i)$, and check it for three letters of the vowel/consonant model by listing the eight probabilities.

The entropy rate is row-then-average on the transition table, and it is the floor for compressing text from the model.

Shannon's Estimates for English

Bits per letter, 27 symbols (Shannon 1951)

model	uniform	monographic	digraphic	trigraphic	long context
bits per letter	4.75	4.03	3.32	3.1	about 1

Each better model of the past lowers the bits needed for the next letter; with a hundred letters of context a human needs about one bit.

The guessing game (the “long context” entry)

Predictor: a person who sees the preceding 100 letters and guesses the next until right. Encoding: $x_1 \dots x_n \mapsto (g_1, \dots, g_n)$, g_i = number of guesses at position i . A decoder with the same predictor recovers the text, so

$$H(\text{English}) \leq H(\text{guess counts}) \in [0.6, 1.3] \text{ bits per letter (Shannon's bounds).}$$

- “Prediction is compression” (Section 6) with a human predictor, twenty-five years before arithmetic coding.
- Redundancy $1 - H / \log 27$: 79% at $H = 1$, 73% at $H = 1.3$; Shannon's stated figure is “roughly 75%”.

Evaluating a Model: Cross-Entropy on Real Text, and Perplexity

Definition

A model q gives a pmf $q(\cdot \mid \text{context}_i)$ for the letter at each position i . Its **cross-entropy on the text** $x_1 \dots x_n$ is

$$\frac{1}{n} \sum_{i=1}^n \log \frac{1}{q(x_i \mid \text{context}_i)} \quad \text{bits per letter,}$$

and its **perplexity** is $PP = 2^{\text{cross-entropy}} \in [1, \infty)$.

The bits per letter the model would need to compress this text; perplexity is the same number as an effective count of equally likely letters.

- On long text from a stationary source P the average tends to the *cross-entropy rate* $\lim_n H(P_n, q_n)/n$ (P_n, q_n : pmfs of n -letter blocks), at least the entropy rate of P by Gibbs.
- A model giving probability 0 to a letter that occurs has infinite cross-entropy; practical models never assign exactly 0.

Perplexities of Shannon's models

Uniform: 4.75 bits, $PP = 27$. Monographic: 4.03, $PP = 16$. Digraphic: 3.32, $PP = 10$, as if choosing among ten equally likely letters. Trigraphic: 3.1, $PP = 8.6$. A model at 1 bit: $PP = 2$.

Back to the Cipher: Frequency Analysis

Theorem (statistics survive a substitution)

Key $k: S \rightarrow S$ a bijection; ciphertext $k(X_1), k(X_2), \dots$. Its monographic and digraphic pmfs are those of the plaintext with the letters relabelled:

$$p_{\text{cipher}}(k(x)) = p(x), \quad p_{2,\text{cipher}}(k(x), k(x')) = p_2(x, x') \quad \text{for all } x, x' \in S.$$

Proof

$\mathbb{P}(k(X_i) = k(x)) = \mathbb{P}(X_i = x)$ since k is injective; likewise for pairs. □

The cipher hides which letter is which; it cannot hide how often each occurs or which follows which.

- The most frequent ciphertext letter is $k(\text{E})$; the ciphertext letter that is always followed by the same letter is $k(\text{Q})$, and what follows it is $k(\text{U})$; the frequent pairs are $k(\text{T})k(\text{H})$, $k(\text{H})k(\text{E})$, $k(\text{I})k(\text{N})$, $k(\text{E})k(\text{R})$.
- Entropies survive too: $H(p_{\text{cipher}}) = H(p)$, and the entropy rate of the ciphertext is that of English. A substitution hides nothing about the redundancy; the next slide turns that into a length.

Back to the Cipher: Unicity Distance

Unicity distance (Shannon)

Redundancy per letter $R = \log 26 - H$; key entropy $H(K)$; the key is determined after about $U = H(K)/R$ letters.

cipher	keys	$H(K)$ bits	U at $H = 1.5$ ($R = 3.2$)	U at $H = 1$ ($R = 3.7$)
substitution	$26!$	88.4	28 letters	24 letters
Caesar shift	26	4.7	1.5	1.3
Vigenère, key length 5	26^5	23.5	7.3	6.4

Each ciphertext letter leaks R bits about the key, so after $H(K)/R$ letters there is no key entropy left.

- $H = 1.5$ is Shannon's working value from short-context estimates on 26 letters; $H = 1$ is the long-context figure of two slides back. The unicity distance is a rough figure either way.
- Below U several keys give plausible English; above it the decryption is unique. Question 3 is answered by two entropies, the key's and the language's.

Exercise

1. A language model reports perplexity 20 per word. How many bits per word is that? If a text has 10^4 words, how large is the compressed file in bytes?
2. An attacker who uses only letter frequencies is, in effect, modelling English monographically, at 4.03 bits per letter on 27 symbols. Compute the redundancy $\log 27 - 4.03$ and the resulting unicity distance for the 88.4-bit substitution key. Why do frequency counts need over a hundred letters when the unicity distance is under thirty?
3. Show that for any model q and any stationary source P , the cross-entropy rate is at least the entropy rate of P , and say exactly which inequality is used and where.
4. Estimate the unicity distance of a substitution cipher on a language whose letters are independent and uniform. What does this say about the security of encrypting compressed text?

Cross-entropy on real text scores a model in bits, and the redundancy of a language is what a cipher leaks per letter.

Answers to the In-Slide Exercises (1)

- **Codes.** (1) 110|0|111|0|10: left, up, right, up, down. 1110010: 7 bits. (2) (1, 2, 2, 3): $\frac{9}{8} > 1$, impossible. (1, 2, 3, 4): 0, 10, 110, 1110, $L = 1.875$. (2, 2, 2, 2): $L = 2$. (1, 3, 3, 3): 0, 100, 101, 110, $L = 2$. (3) 0, 10, 11: $L = 1.5$. (4) Not prefix-free (0 begins 01), but decodable: a new codeword starts at every 0. $\sum 2^{-\ell} = \frac{15}{16}$; 0, 10, 110, 1110. (5) $6 \cdot \frac{1}{4} > 1$. 00, 01, 100, 101, 110, 111: $L = \frac{16}{6} = 2.667$. Not 2.5: $\bar{H} = \log 6 = 2.585$ is a lower bound (Section 4).
- **Information.** (1) 4 bits; 19.3 bits; $2^{-10} \approx 0.001$. (2) 1; $\log 3 = 1.585$; $\log 6 = 2.585$ bits. (3) 2.585, 2.585, $\log 36 = 5.170$: the events $\{Y = 1\}, \{X = 7\}$ are independent, so the informations add. $I(\{X = 2\}) = 5.170 = I(\{Y = 1\} \cap \{X = 2\})$: $\{X = 2\} \subseteq \{Y = 1\}$, no independence, no additivity. (4) $f(1) = f(1) + f(1)$; $f(\frac{1}{8}) = 3f(\frac{1}{2})$; $2f(1/\sqrt{2}) = f(\frac{1}{2})$. (5) Exactly when $p_X(x) = 2^{-\ell(x)}$ for all x : if every bit is a fair flip, a codeword of length ℓ is emitted with probability $2^{-\ell}$.
- **Entropy.** (1) Four bars of width $\frac{1}{4}$, height 2: $H = 2$. Widths $\frac{1}{2}, \frac{1}{4}, \frac{1}{4}$, heights 1, 2, 2: $H = 1.5$. (2) $H(W) = 1$; V has pmf $(\frac{1}{6}, \frac{1}{6}, \frac{4}{6})$, $H(V) = 1.252$. (3) Sum 3.274, maximum (pmf (1, 3, 5, 7, 9, 11)/36) 2.320 bits; the maximum is cheaper. (4) $p \log \frac{1}{p} \geq 0$, $= 0$ iff $p \in \{0, 1\}$; all terms vanish iff one $p_x = 1$. (5) Pairs (0.81, 0.09, 0.09, 0.01) with code 0, 10, 110, 111: 1.29 bits per pair, 0.645 per symbol.

Answers to the In-Slide Exercises (2)

- **Source coding.** (1) No: $2.5 < 2.585 = H$. (2) $a = (2^{-\ell_1}, \dots)$, Kraft, then Gibbs with $p = (\frac{1}{2}, \frac{1}{4}, \frac{1}{8}, \frac{1}{8})$: $L \geq 1.75$. (3) $L^* = 1$ is consistent with $L^* < 1.469$. Pair lengths $\lceil \log \frac{1}{0.81} \rceil, \dots = 1, 4, 4, 7$: 1.60 bits per pair, 0.80 per symbol. (4) $0.469 + \frac{1}{k} < 0.6$ needs $k \geq 8$; the lower bound H holds for every k . (5) Then $\ell(x) = \log \frac{1}{p_X(x)}$ exactly, $\sum 2^{-\ell} = 1$ and $L = H$; each internal node halves the probability beneath it.
- **Cross-entropy.** (1) $\frac{1}{4}(1 + 2 + 3 + 3) = 2.25$; $\frac{1}{4} \cdot 1 + \frac{1}{2} \cdot 2 + \frac{1}{8} \cdot 3 + \frac{1}{8} \cdot 3 = 2.0$. (2) Every length is $\log n$. Conversely, if $\sum_x P(x)\ell(x)$ is constant in P , then P concentrated on x , then on x' , gives $\ell(x) = \ell(x')$; equal lengths with Kraft sum 1 force $\ell = \log n$. (3) $H(P, Q) = 1.2075$ (fair coin, $\frac{1}{4}/\frac{3}{4}$ code); $H(Q, P) = 1$ ($\frac{1}{4}/\frac{3}{4}$ coin, fair code). (4) $a = Q$, $\sum Q = 1$. (5) Cost $p \log \frac{1}{q} + (1-p) \log \frac{1}{1-q} = H((p, 1-p), (q, 1-q))$, minimised at $q = p$ by Gibbs, value $h(p)$.
- **Relative entropy.** (1) $D(P\|Q) = 1 - 0.469 = 0.531$; $D(Q\|P) = 1.737 - 1 = 0.737$. (2) 0.2075 and 0.1887: the fair coin sends its 2-bit codeword half the time instead of a quarter. (3) $\sum P \log(nP) = \log n - H(P) \geq 0$; maximum $\log n$, at the uniform pmf only. (4) Uniform code: cost 2 under either truth, waste 0.25 and 0.25; the old code wastes 0 and 0.875. (5) Relabelling permutes the terms of the sum. $P = (1 - 10^{-12}, 10^{-12})$, $Q = (0.95, 0.05)$: $D(P\|Q) = 0.074$, $D(Q\|P) = 1.71$.
- **Log loss.** (1) 1, 0.469, 0.677 bits; 0.9 is honest; at 0.99 the 10% tail costs $\log 100 = 6.64$ bits each time. (2) Cost $0.3 \log \frac{1}{q} + 0.7 \log \frac{1}{1-q}$, minimised at $q = 0.3$ (Gibbs), value $h(0.3) = 0.881$; hedging with 0.5 costs 1. (3) Labels with $H(p) = 0.5$ bits of noise and a perfect model; or noiseless labels and $D = 0.5$ still to remove; the curve alone cannot tell. (4) $q_y = 1 + \lambda/(2p_y)$, $\lambda = -4/\sum 1/p_y = -0.387$: $q = (0.613, 0.355, 0.032)$; penalty 0.387 there, 0.400 at $q = p$. (5) With $q_2 = 1 - q_1$, $p(1-q)^2 + (1-p)q^2$ has derivative $-2p(1-q) + 2(1-p)q$, zero at $q = p$.

Answers to the In-Slide Exercises (3)

- **Wordle.** (1) OF: buckets {AM,IN}, {NO,TO}, {OF}, {ON}: 1.918 bits. AM: (5, 1), 0.650. (2) OF and TO (also NO, ON, from outside the reduced set {AM,OF,TO}) give three distinct patterns: $\log 3 = 1.585$ bits; AM gives (2, 1), 0.918. (3) Π has at most N non-empty buckets and at most 243 values; $H \leq \log(\text{number of values})$, equality iff uniform. $\log 243 = 7.92$ bites at the start, $\log N$ once $N < 243$. (4) Removing g removes a bucket of size 1 and changes N ; a guess outside W is not obliged to spend a pattern on itself and can split W more evenly. (5) The score depends on whether the guess can win now and on how each bucket resolves later, not only on bucket sizes; you need the game tree below each bucket.
- **Joint and conditional entropy.** (1) Each row: Z uniform, $\log 6$; $H(Z | Y) = \log 6 = H(Z)$: every row has the table's distribution, $Z \perp\!\!\!\perp Y$. (2) $\log n_x$: 0, 1, 1.585, 2, 2.322, 2.585, 2.322, 2, 1.585, 1, 0; weighted by $n_x/36$: 1.896. (3) Row y : $M = y$ on y cells, $M = z$ on one cell each for $z > y$. $H(M | Y = y) = 2.585, 2.252, 1.792, 1.252, 0.650, 0$; $H(M | Y) = 1.422$; $H(M) = 2.320$; $H(Y | M) = 2.585 + 1.422 - 2.320 = 1.687$. (4) $H(X, g(X)) = H(X) + H(g(X) | X) = H(X)$ and $= H(g(X)) + H(X | g(X)) \geq H(g(X))$; equality iff $H(X | g(X)) = 0$, i.e. g injective on S_X . (5) Cells $\frac{1-\varepsilon}{2}$ on the diagonal, $\frac{\varepsilon}{2}$ off it. $H(Y | X = x) = h(\varepsilon)$, $H(Y | X) = h(\varepsilon)$, $H(X, Y) = 1 + h(\varepsilon)$, $H(Y) = 1$.

Answers to the In-Slide Exercises (4)

- **Mutual information.** (1) $I(Y; Z) = 0$ (independent); $I(X; Z) = 0.689$ by symmetry; $I(X; (Y, Z)) = H(X) - 0 = 3.274$. (2) Independent bits: $2 \cdot 0.531 = 1.062$ bits. Same bit twice: rows $(0, 0), (1, 1)$ have probability 0.41 each and $H(X | \cdot) = h(0.81/0.82) = 0.094$; rows $(0, 1), (1, 0)$ have 0.09 each and entropy 1. $H(X | Y_1, Y_2) = 0.258$, $I = 0.742$: less than two independent uses, more than one. (3) $I = H(Y)$ iff $H(Y | X) = 0$ iff every row $\{X = x\}$ has one value of Y . Both iff each is a function of the other. (4) $H(Y | X) = 1$; Y has pmf $(1, 2, 2, 2, 2, 2, 1)/12$, $H(Y) = 2.752$; $I = 1.752 < 2.585$. Given $Y = 4$, say, X is 3 or 4: the noise hides one bit's worth of X almost everywhere. (5) $I(A; \Pi) = H(\Pi) \leq \log 243$ since Π has 243 values; $I \leq H(A) = \log N$. The first: $7.92 < 11.18$.
- **Markov models and entropy rate.** (1) $\sum_x \pi(x)Q(x, x') = \pi(x') \sum_x \pi(x) = \pi(x')$; every row entropy is $H(\pi)$, so the average is $H(\pi)$; $\mathbb{P}(X_{i+1} = x' | X_i = x) = \pi(x')$ does not depend on x : the monographic model. (2) Markov: $p(Q)Q(Q, U) \approx p(Q)$; monographic: $p(Q)p(U) \approx 0.03p(Q)$, about 33 times smaller. (3) $\pi(V) \cdot 0.8 = \pi(C) \cdot 0.9$: $\pi = (\frac{9}{17}, \frac{8}{17})$; rate $\frac{9}{17}h(0.2) + \frac{8}{17}h(0.9) = 0.603$ bits, against 0.893 before and $h(\frac{9}{17}) = 0.998$ for its own monographic model: much more predictable, since letters now nearly alternate. (4) $H(X_1, \dots, X_k) = H(X_1, \dots, X_{k-1}) + H(X_k | X_1, \dots, X_{k-1})$ is the two-variable rule for the pair $((X_1, \dots, X_{k-1}), X_k)$; sum over k . Markov replaces each term by $H(X_k | X_{k-1}) =$ the rate, for $k \geq 2$. Vowel/consonant, $n = 3$: $0.961 + 2 \cdot 0.893 = 2.747$, and the eight triple probabilities $\pi(a)Q(a, b)Q(b, c)$ give 2.747 directly.

Answers to the In-Slide Exercises (5)

- **Evaluating models, redundancy, ciphers.** (1) $\log 20 = 4.32$ bits per word; $4.32 \cdot 10^4$ bits ≈ 5.4 kB. (2) $R = 4.755 - 4.03 = 0.725$ bits per letter; $U = 88.4/0.725 \approx 122$ letters. Frequency counts use only the monographic redundancy, 0.7 of the 3.7 bits per letter that English actually leaks; the rest is in digraphs, words and grammar, which a human decrypter uses and a letter count does not. (3) On a block of n letters, $H(P_n, q_n) \geq H(P_n)$ by Gibbs ($a = q_n$, a pmf, so $\sum a = 1$); divide by n and let $n \rightarrow \infty$; the right side tends to the entropy rate. (4) Redundancy $\log 26 - \log 26 = 0$: unicity distance infinite, the ciphertext never determines the key. Compressed text has almost no redundancy, so encrypting it is far harder to break by statistics; compress first, then encrypt.